

④ Create a structure variable student and estimate the size of structure manually then write a c program to find size of the structure using size of operator now observe both the outputs are they same? or difference justify your answer.

Ans

Size of structure

```
#include <stdio.h>
```

```
struct student {
```

```
    int roll-num;
```

```
    char grade;
```

```
    float marks; };
```

```
int main()
```

```
{ struct student s1;
```

```
    printf("Size of structure student = %zu bytes.\n",  
           sizeof(struct student));
```

```
    printf("Size of s1 = %zu bytes\n", sizeof(s1));
```

```
    return 0; }
```

Output: size of structure student is 12 bytes
size of s1 = 12 bytes

Manually

int $\rightarrow$ 4 bytes

char $\rightarrow$ 1 bytes

float $\rightarrow$ 4 bytes

total = 9 bytes

Due to structure padding